

Introduction to Computer Science

Spring 2026

hgl235@nyu.edu

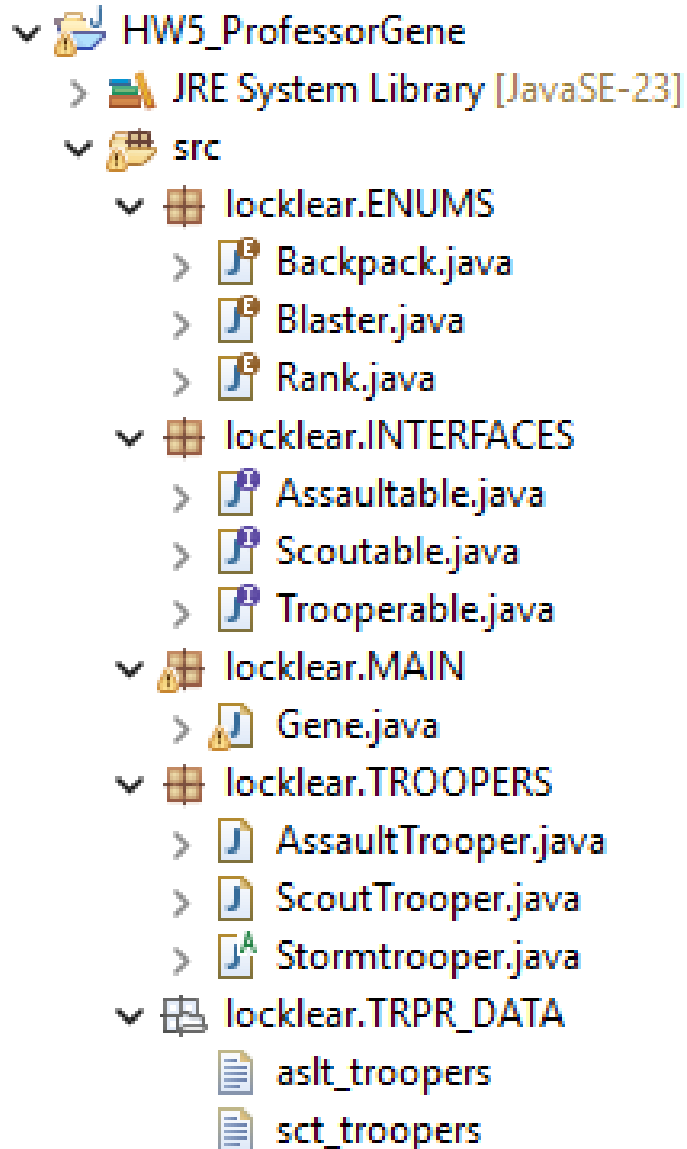


Homework 5

OOP Persistence

Professor HG Locklear

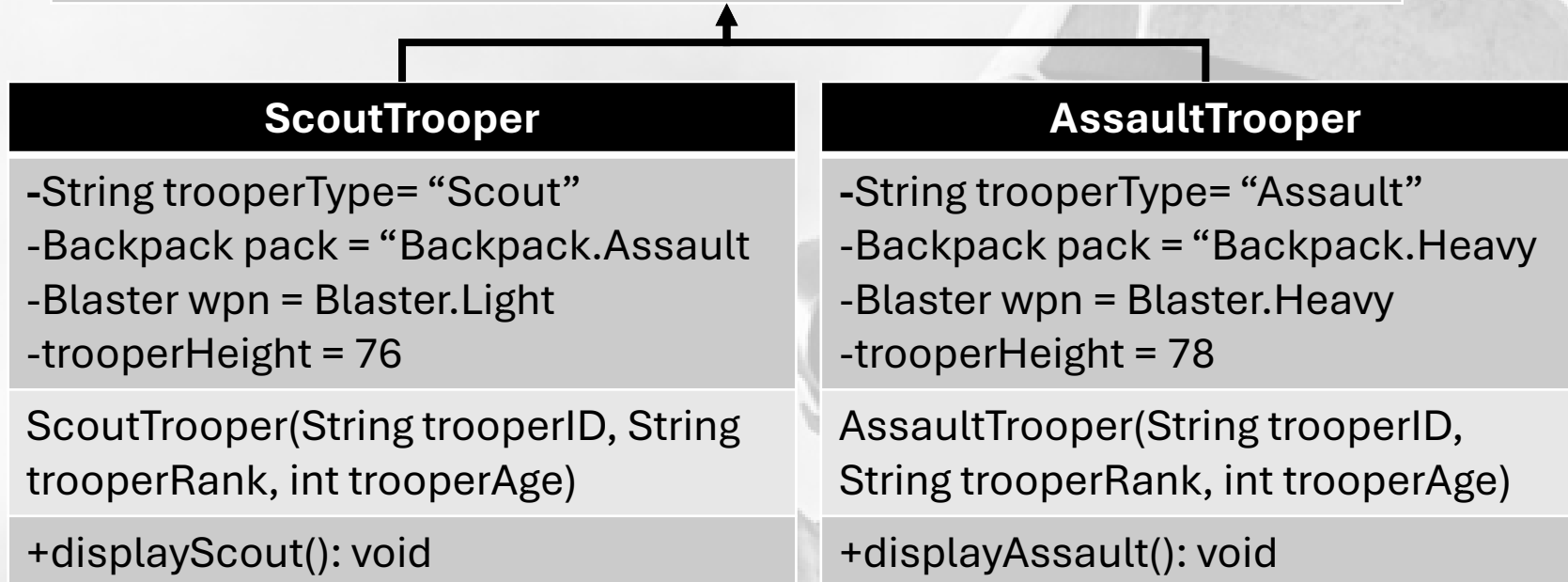
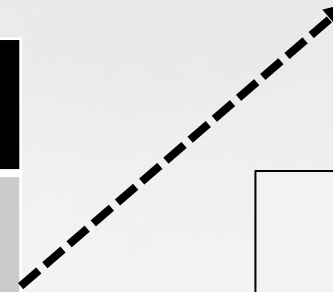
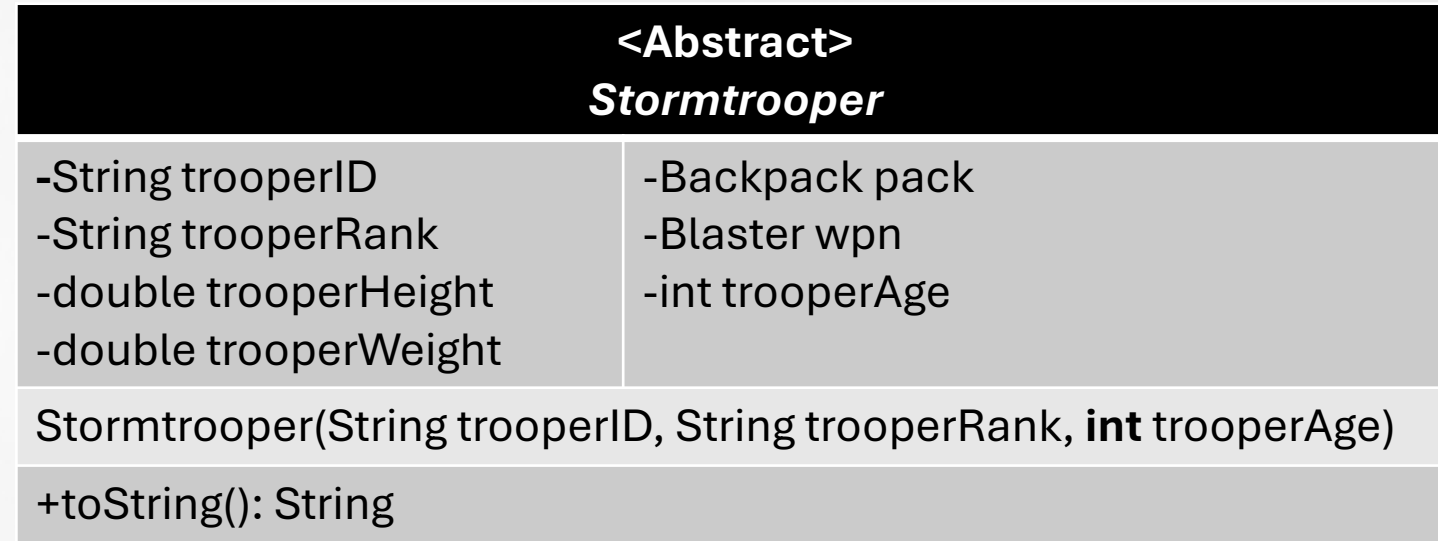
Program Structure



Create your Project, Program Structure, and main method **EXACTLY** as shown
YOU MUST replace my name with yours where appropriate.

```
public class Gene {  
  
    public static void main(String[] args) throws IOException, ClassNotFoundException {  
  
        Trooperable.storeTroopers(Trooperable.buildTroopers(2,2));  
        ArrayList<Stormtrooper> trp = Trooperable.retrieveTroopers();  
        System.out.println();  
  
        for(Stormtrooper s: trp) {  
            if(s.getClass().toString().contains("Scout")) {  
                ((ScoutTrooper)s).displayScout();  
            }  
            if(s.getClass().toString().contains("Assault")) {  
                ((AssaultTrooper)s).displayAssault();  
            }  
  
            System.out.println();  
        }  
    }  
}
```

Stormtrooper Class



- Special Instructions**
- Subclasses **must use** superclass constructor
 - All Troopers have a randomly generated trooper ID that is formatted as **\$\$-####** (\$ = Capital Letter # = Digit)
 - **\$** = A,B,C,D, or E for ScoutTroopers and **\$** = G,H, I,J,K, or L for AssaultTroopers
 - All Troopers have a randomly generated age between 24 and 28 inclusive.
 - All Troopers have a randomly generated weight between 200 and 215 pounds inclusive.

Enumerations

<Enum> Backpack

Assault("Aslt",50)
Heavy("Hvy",100)

-String type
-int capacity

-Backpack(String type, int capacity)

<Enum> Blaster

Light("Lt",150)
Heavy("Hvy",50)

-String type
-int rounds

-Blaster(String type, int rounds)

<Enum> Rank

SGT,CPL,PVT



Class Methods

Method Specifications			
Class/Method	Input	Processing	Output
Stormtrooper toString	None	<u>Returns</u> a String representation of the Stormtrooper object in any format	String
ScoutTrooper displayScout	None	<u>Prints to the console</u> a String representation of the ScoutTrooper object in the format shown on the Expected Output slide	void
AssaultTrooper displayAssault	None	<u>Prints to the console</u> a String representation of the AssaultTrooper object in the format shown on the Expected Output slide	void



Trooperable Interface

<<Trooperable>>

+**static** generateTrooper(String trprType): Stormtrooper
+**static** ArrayList<Stormtrooper> buildTroopers(int scouts, int assaults)
+**static** storeTroopers(ArrayList<Stormtrooper> trprs): void
+**static** retrieveTroopers(): ArrayList<Stormtrooper>



Method Specifications			
Method	Input	Processing	Output
generateTrooper	None	<u>Returns</u> the specified Stormtrooper object by randomly generating the trooperID, trooperRank, trooperWeight, and trooperAge	Stormtrooper
buildTrooper	scouts assaults	<u>Returns</u> an ArrayList of the specified number and type of Stormtrooper objects specified in the parameters	ArrayList
storeTroopers	trprs	<u>Serializes</u> the Stormtrooper objects into the sct_troopers or aslt_troopers file from the trprs ArrayList based on the Stormtrooper subclass	Binary File
retrieveTroopers	None	<u>Deserializes</u> the Stormtrooper objects in the sct_troopers and aslt_troopers files and returns them in a single ArrayList. <u>Prints to the Console</u> Scout Trooper Deserialized and Assault Trooper Deserialized	ArrayList

Interfaces

<<Scoutable>>

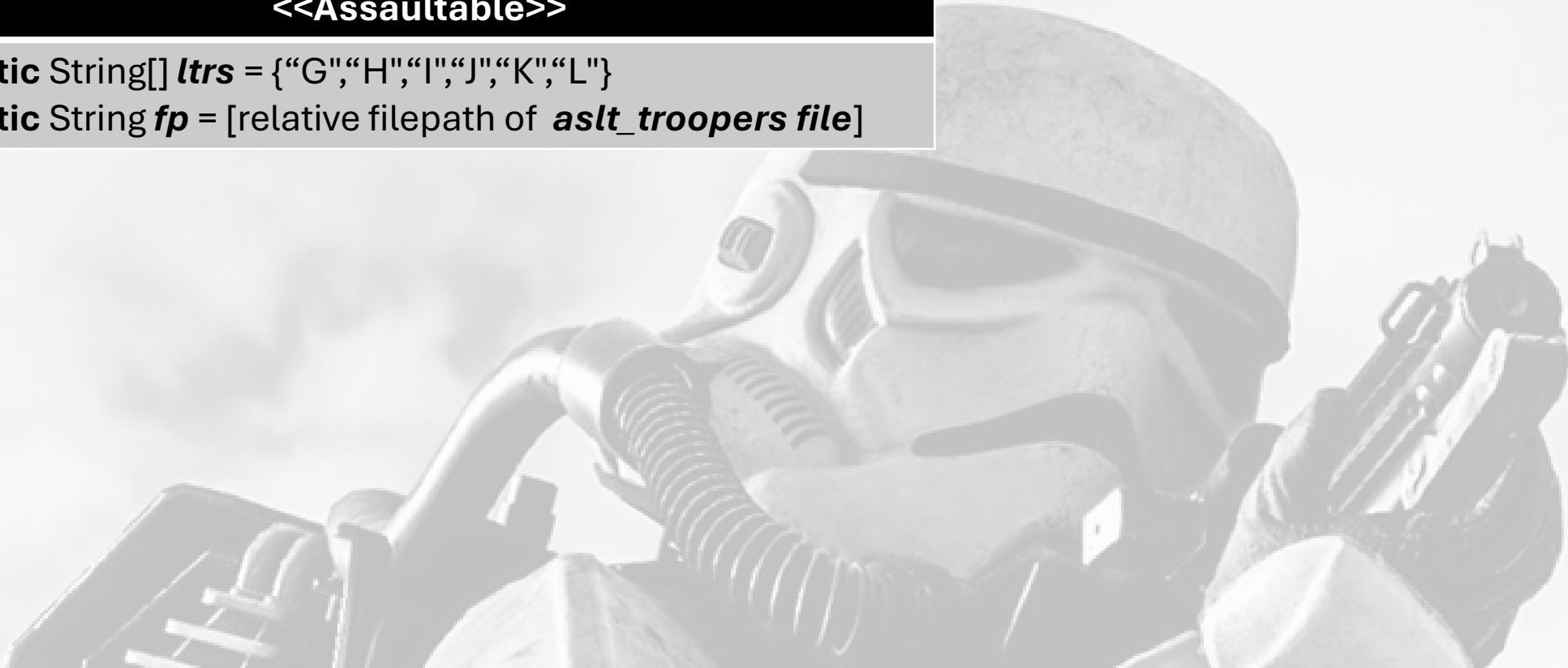
+static String[] ***ltrs*** = {"A","B","C","D","E","F"}

+static String ***fp*** = [relative filepath ***of sct_troopers file***]

<<Assaultable>>

+static String[] ***ltrs*** = {"G","H","I","J","K","L"}

+static String ***fp*** = [relative filepath of ***aslt_troopers file***]



Expected Output

console

```
Scout Troopers Deserialized
Assault Troopers Deserialized

Scout Trooper
Trooper ID: CE-1985
Rank: CPL      Height: 76.0    Weight: 207.0    Age: 24
Pack: Assault  Pack Capacity: 50      Weapon: Light  Wpn Rounds: 150

Scout Trooper
Trooper ID: DB-7042
Rank: CPL      Height: 76.0    Weight: 205.0    Age: 28
Pack: Assault  Pack Capacity: 50      Weapon: Light  Wpn Rounds: 150

Assault Trooper
Trooper ID: HI-2853
Rank: PVT      Height: 78.0    Weight: 215.0    Age: 25
Pack: Heavy    Pack Capacity: 100  Weapon: Heavy  Wpn Rounds: 50

Assault Trooper
Trooper ID: JK-8059
Rank: PVT      Height: 78.0    Weight: 203.0    Age: 28
Pack: Heavy    Pack Capacity: 100  Weapon: Heavy  Wpn Rounds: 50
```

sct_troopers

```
sct_troopers x
1 1 1 1 sr 1 locklear.TROOPERS.ScoutTrooper
2 trooperAgeD
3 trooperHeightD
4 trooperWeightL 1 packt 1 1 locklear/ENUM
5 q 1 ~ 1 1
```

aslt_troopers

```
aslt_troopers x
1 1 1 1 sr 1 locklear.TROOPERS.AssaultTrooper
2 trooperAgeD
3 trooperHeightD
4 trooperWeightL 1 packt 1 1 locklear/ENUMS/B
5 q 1 ~ 1 1
```